

Monitoring the DAQ Performance of the sPHENIX Experiment

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Abstract

The sPHENIX experiment [1] is an upgrade of the existing PHENIX apparatus at the Relativistic Heavy-Ion Collider (RHIC). While taking data, the shift crews operate the detector 24/7, and periodically review monitoring information to ensure that the data coming from each detector component is of high quality. There are many aspects of the data that need monitoring, and a balance has to be found between comprehensive monitoring of a given detector, and the ability of the shift crew to digest what is being presented. Consequently, there are two aspects to the monitoring: obtaining the data to be monitored, and presenting them to the shift crews in a concise, and easy to interpret way. In this project the monitoring displays for the data acquisition system were improved in order to ensure that the readout electronics chains exhibit readout errors at an acceptable rate. Specifically, monitoring displays for packet size- the amount of data that a given detector component contributes to the overall data stream- and the bunch crossing number – which ensures the trigger and timing system is properly locked on to the RHIC beam rotation – were created.

1 Introduction

During the summer, three different projects were completed, two related to the monitoring of the sPHENIX data acquisition system and the trigger and timing system, and one to find an answer to a small mystery that had been observed a year earlier.

1.1 The sPHENIX Data Acquisition System

The sPHENIX data acquisition system acquires data from 9 different detector systems with about 60 different detector elements that are read out by individual instances of a common data acquisition system called RCDAQ. The resulting 60 data streams combined provide the data from individual RHIC beam collisions, of which the experiment acquires and stores about 15,000 per second [2].

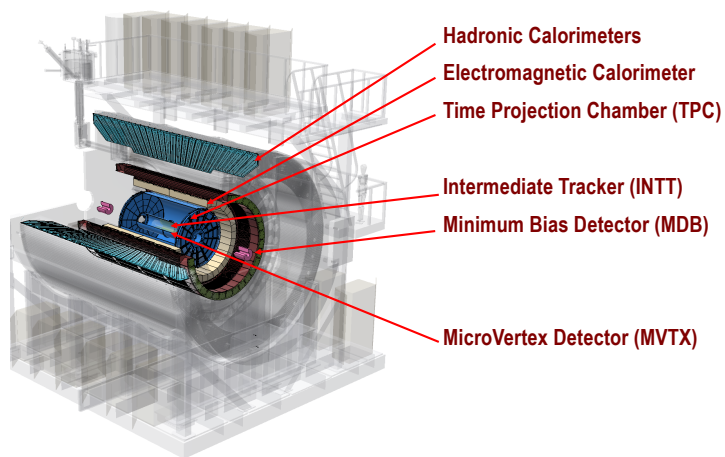


Figure 1: View of the sPHENIX detector with its component subdetectors.

The online monitoring system is designed to find most problems that affect the physics performance on a time scale of 15 minutes, in some cases less. The information reviewed here is generally different from conditions that need immediate attention and generate alarms, such as tripped power supplies, cooling water leaks, or over-temperatures in the detector electronics. Instead, the information presented here is typically about the proper functioning of a detector component from a physics data quality point of view. For example, one needs to know that all components operate at the expected gain, that all channels give signals and that there are no "dead" regions, and (for select detectors) certain physics quantities that can be measured quickly are in the expected range.

Most detectors' online monitoring only requires a relatively small subset of the overall information collected in the about 60 individual data streams. For example, both the Inner and Outer Hadronic Calorimeter each require access to just two data streams that hold the information for those detectors. Conversely, monitoring the data acquisition itself requires access to all its components. Each detector system has a data volume budget that it cannot

routinely exceed without slowing down the entire data acquisition, or taxing the storage system beyond its design limits. Reasons for exceeding this budget can be a faulty component that injects noise into this system, or insufficient zero-suppression of the data, which can dramatically increase the data volume. This can in turn be caused by wrong or no calibration constants loaded into the readout system, but can have many other causes as well, making online monitoring essential for the smooth running of the experiment.

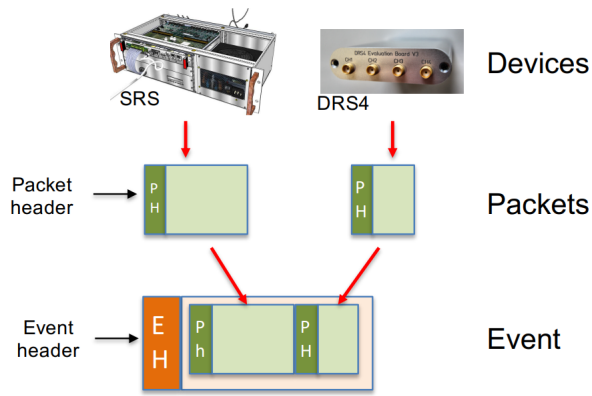


Figure 2: The hierarchy of data that are also used to process the online monitoring data. stored in a file. For easier management, each detector is assigned a number that determines the packet id range (ex: EmCal is detector #6, so packet ranges are in the 6000 range) [3].

1.2 The sPHENIX Trigger and Timing system

One of the most important components of the wider sPHENIX data acquisition system is the Trigger and Timing system, referred to as GL1/GTM (“global level 1/granule timing module”), which distributes the timing information to the front-end electronics, as well as trigger information. Among the most visible information is the proper synchronization of the timing system with the rotation of the beam in RHIC. The two beam lines have room for 120 “bunches” each that can collide at various interaction points, of which IP8, where the sPHENIX experiment is located, is one. Each ring gets filled with up to 111 bunches. The remaining bunch slots need to remain empty in order to be able to dump the beam properly. On the ring, this abort gap defines the front of the bunch train and the end of the train. At a particular place, a set of powerful magnets can “derail” the beam into a dedicated beam dump, and they do that once the end has passed by and

before the front arrives, that is, while the abort gap is at that switch. This abort gap plays an important role for the calibration procedures of the experiment, too. The 9 empty bunches constitute beam crossings where no collisions can take place, and this is the time when LED light and other calibration pulses can be injected into the otherwise empty detector. For this to work, the timing system must be properly synchronized, and this synchronization must be carefully monitored.

1.3 Determining Muon Flux along the sPHENIX Beamline

In order to augment the largely code-centric work described above with an additional component that emphasizes experimental techniques, we set up a muon telescope in the sPHENIX “Assembly Hall”, right outside of the shield wall that protects humans from ionizing radiation. This was an attempt to answer a year-old question that had caused some discussion among sPHENIX scientists.

2 Methods

2.1 EmCal and DAQ Monitoring

There are two main aspects of the online monitoring process: the analysis and preparation of the data in question from the detector data streams, and the proper delivery of this information to the shift operators. Therefore, a server that processes events and client that can be accessed to display histograms were used to implement two new monitors. The first of these monitors the average size of the packets of a given detector, and the second monitors “check sum errors”.

Currently the data from sPHENIX is not zero-suppressed, so all packets for a given id come out with the same/similar size. An example of these packets from the emCal is shown below.

```
dlist -n 10 /bbox/commissioning/emcal/beam/beam_seb07
-00023746-0000.prdf | grep 6025
Packet 6025 5981 0 (Unformatted) 93 (IDDIGITIZER_31S)
Packet 6025 5981 0 (Unformatted) 93 (IDDIGITIZER_31S)
Packet 6025 5981 0 (Unformatted) 93 (IDDIGITIZER_31S)
```

In order to determine trends in the packets, a “running mean” of the values was calculated. This was done by taking the last n values of the time series of packets, and calculating the mean. For this monitor, the last 50 packets were used, adding the latest value, recalculating, and dropping

what is then the 51st value. In this way it is easy to spot a trend that would else be buried in too many fluctuations. Calculating this full running mean for large n can lead to an excessive amount of memory allocated, so a mathematically similar approximation for the value – the pseudo-running-mean – was used. The example used to develop this monitor was a file with 128 packets, so a histogram with 128 bins was created and filled with the pseudo running mean of that packet size there. The chosen depth of this approximation was the last 50 values.

The pseudo running mean will work very well for monitoring values which do not vary too much, and makes it very well-suited for gain monitoring and similar applications, where there are no dramatic changes in the values under normal conditions. It will calculate the new running mean value r_n from the existing value r_o as

$$r_n = (r_o * (n - 1) / n + x) / n \quad (1)$$

where x is the new value in the running mean calculation. If there have been less than n readings so far, it will return the actual mean value.

Since there are often dozens or even hundreds of such displays to review, the information must be simple to understand, concise, and have references to the action to be taken in case of a problem. In other words, if a display shows any aspect or component is not “green”, there is more detailed information to be accessed and acted on. In this particular case of monitoring data packet checksum errors, the packets maintain two different checksums, one value for odd- and one value for even-numbered data channels. This is due to the way the readout electronics is designed. Consequently, there are three possible states we want to indicate: Both checksum values are OK, or a wrong even checksum, a wrong odd checksum. The case that both checksums are wrong is treated as if they individually wrong. A histogram with three bins was created where each bin was filled if one of the three possible conditions was found. For example, the first bin of the histogram was filled and colored green if both the even and odd check sums were OK.

The second bin was filled if the even check sum was wrong, and colored red. Because so few wrong values occur, the scale was set to logarithmic. In this was it becomes easy for the shift crew to determine if the data is of good quality.

2.2 Trigger Monitoring

In terms of monitoring the trigger, a histogram with 120 bins to correspond to the 120 bunches each that can collide at various interaction points was created. From bunch 111, there are 9 empty bunches which constitute

beam crossings where no collisions can take place, and therefore the amount of entries in the corresponding bins of the histogram should be largely empty (a few calibration events and other random triggers are found here, but are orders of magnitude suppressed).

3 Results

3.1 Online Monitoring for the Trigger and Timing System

Figure 3 plots the 120 bins (corresponding to the 120 bunches), where nine of those bins- from 111-120 are be largely empty, demonstrating the proper location of the abort gap.

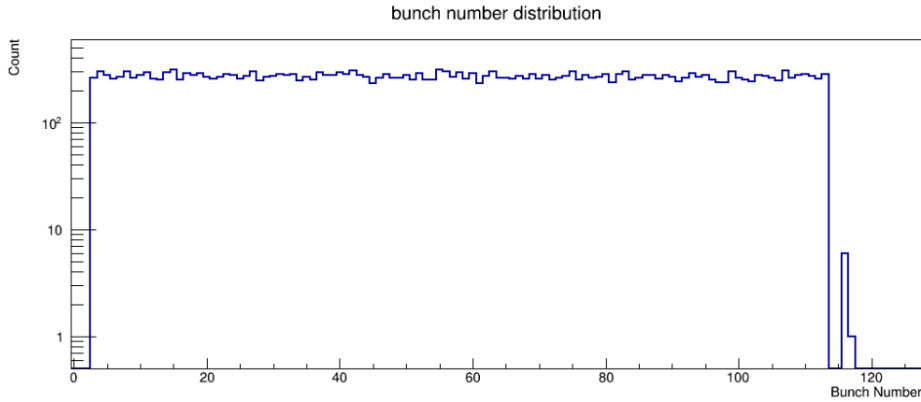


Figure 3: Trigger Plot

3.2 Online Monitoring for the Inner and Outer Hadronic Calorimeters

In order to determine trends in the packets, a "pseudo running mean" of the values was calculated. Figure 4 demonstrates this by plotting these values for the electromagnetic calorimeter.

Figure 5 illustrates the three possible values for the check sum, with the majority being good. For the shift crew, seeing a disproportionate number of wrong check sums would be a flag to indicate the data being taken is poor quality.

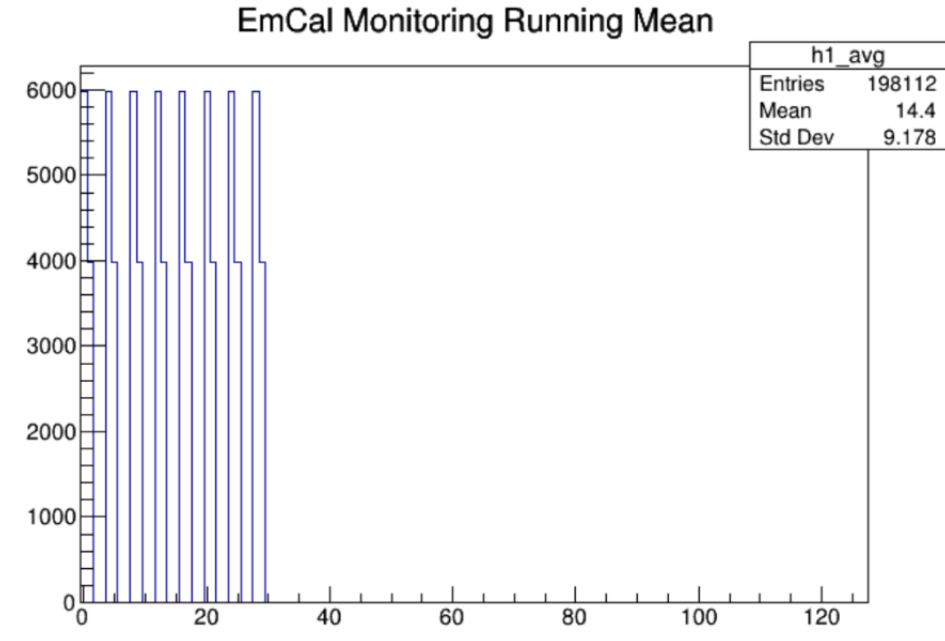


Figure 4: Bunch Number Plot

4 Muon Detection

As described above, we set up a “horizontal” muon telescope in the sPHENIX “Assembly Hall”, outside of the shield wall. Most muons, be it cosmic muons or those created by the RHIC beams, can still penetrate the shielding.

The history of this measurement goes back to the summer of 2022, when the not yet fully assembled sPHENIX apparatus was still in the Assembly Hall. The shield wall was closed while RHIC was in operation for the then-sole STAR experiment. During that time, another muon telescope was set up that served as a trigger for a section of the outer Hadron Calorimeter to be read out to test the proper functioning of the detector, as well as the readout electronics. As a surprise, many more muons than expected were measured at the time, to the point that single muon signals rivaled the yield triggered with the telescope. Only later it was recognized that the increase in muon yield *might* come from additional muons created by beam-gas or other interactions of the RHIC beams, and explain the observed excess. At that time, however, the RHIC operations had ended for the year, and a

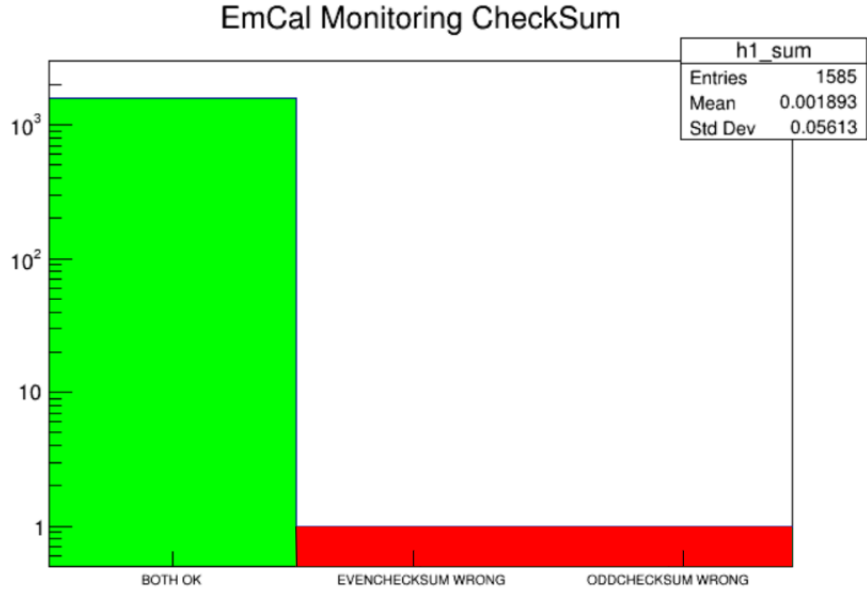


Figure 5: CheckSum Error Plot

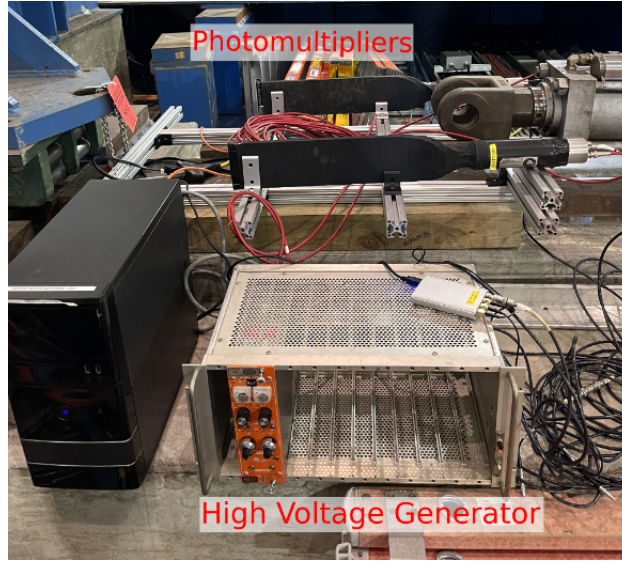


Figure 6: Trigger Plot

definitive answer was impossible to get.

This year, with RHIC providing collisions for both STAR and sPHENIX, we assembled a “horizontal” muon telescope consisting of two scintillator pad-

dles read out with two photomultipliers. The photomultiplier signals were digitized with a “DRS4” Evaluation Board, a waveform sampling device that acts in many aspects like a digital oscilloscope and can sample pulses with a sample rate of up to 5 Billion Samples/s. The DRS4 was read out with the RCDAQ data acquisition system, the same that is in use as the main sPHENIX DAQ system. The device was set up to digitize both photomultiplier signals, but trigger on one of them (the scintillator closer to the camera in the Figure 6). In this way, the setup did not need additional electronics such as discriminators, or gate generators.

4.1 Cosmics Rate and Detection

Figure 7 shows a typical waveform captured on the “trigger channel” 0. If a muon hits the second paddle rather than the first there will be no signal, as the trigger is on channel 0.

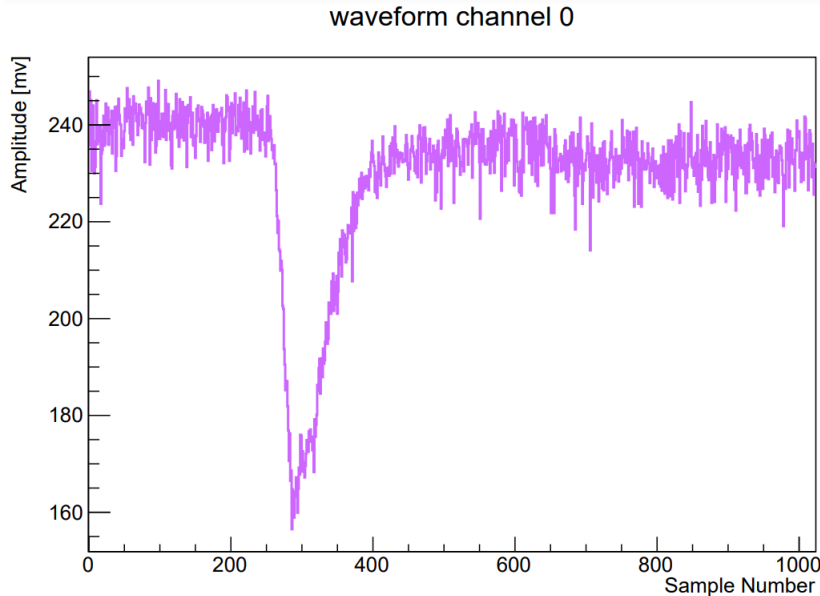


Figure 7: Muon Waveform

Additionally, Figure 8 demonstrates the correlation between beam intensity and muon rate. The upper plot shows the number of observed muons as a function of time in 15 minute bins, while the lower plot shows the beam intensity as measured by the sPHENIX Zero Degree Calorimeter. A 2D histogram, seen in Figure 9 was created to directly compare these intensities, and a slight upward trend is seen, with the bins in the upper right corner representing the muon peaks seen in the previous plot.

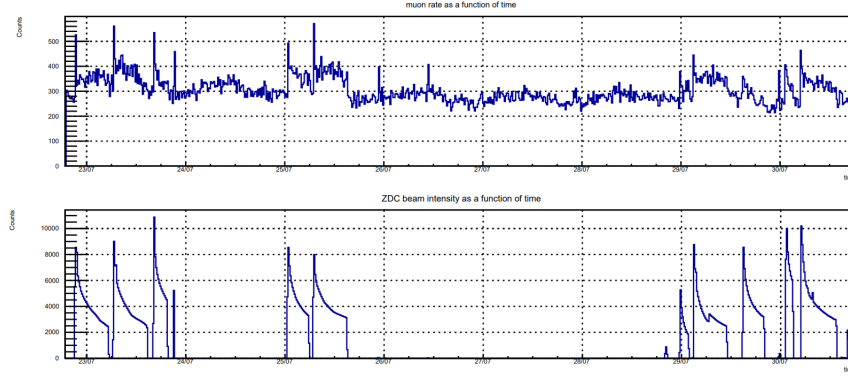


Figure 8: Muon Rate vs Beam Intensity

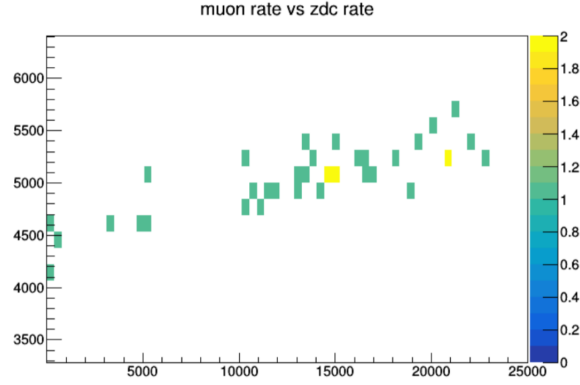


Figure 9: Rate Comparison

Furthermore, Figure 10 demonstrates the potential discriminating between muons hitting the left vs right paddle first. Since the scintillators are an estimated 3ns apart, a left-right muon and a right-left one would be 4ns, or 20 samples, apart. The process for seeing if there is a way to discriminate between these signals involves inverting the signal with an added delay and subtracting the signals to create a new waveform, the bin at which this waveform crosses zero being of interest. This is depicted in Figure 11. The expected difference in timing between right-left and left-right muons was difficult to tell, as seen in Figure 12, the expected "two humps"- one hump for each signal, are not visible with this resolution.

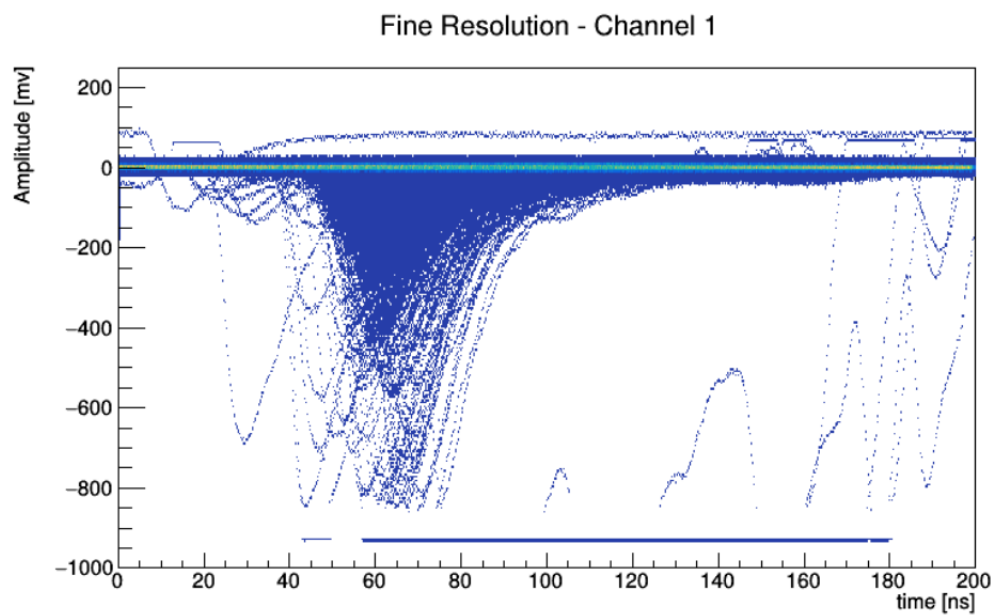


Figure 10: Fine Resolution Plot

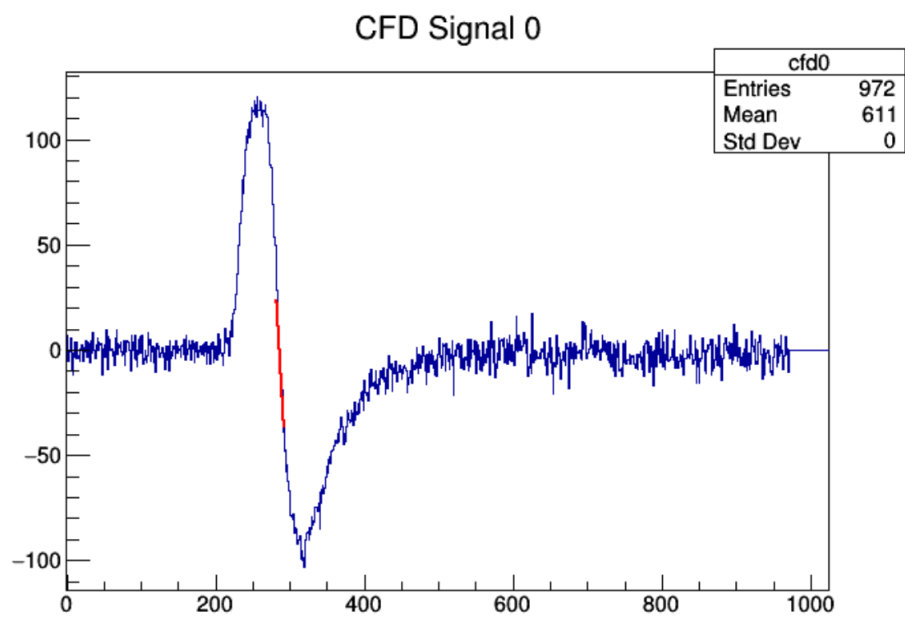


Figure 11: CFD Zero Crossing

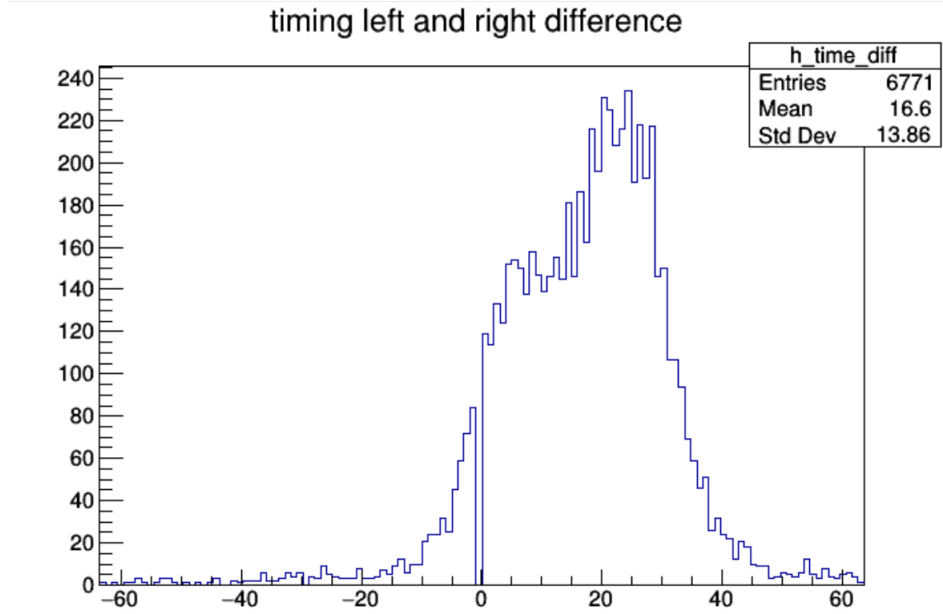


Figure 12: Left-Right Discrimination

5 Conclusions

With this project, we contributed two areas of the online monitoring setup of the sPHENIX experiment, the DAQ, and trigger and timing system. We monitored the packet sizes that have the potential to slow down the DAQ when too large. We also monitored the bunch number alignment of the timing system to line up with the RHIC rotation. Secondly, despite two definitive peaks for right-left and left-right muons not being seen, the correlation between muon rate and beam intensity extremely convincing.

6 Acknowledgments

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References

- [1] A. Adare et al. An Upgrade Proposal from the PHENIX Collaboration. 2014.
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